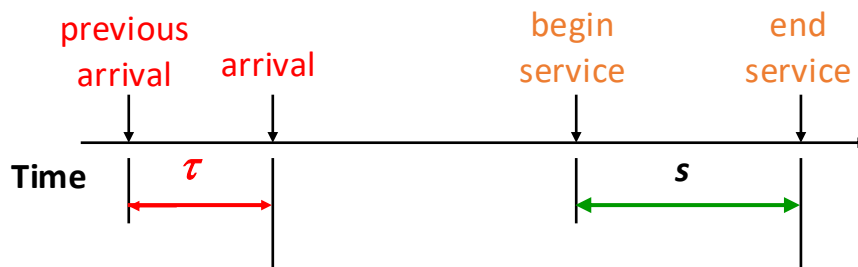
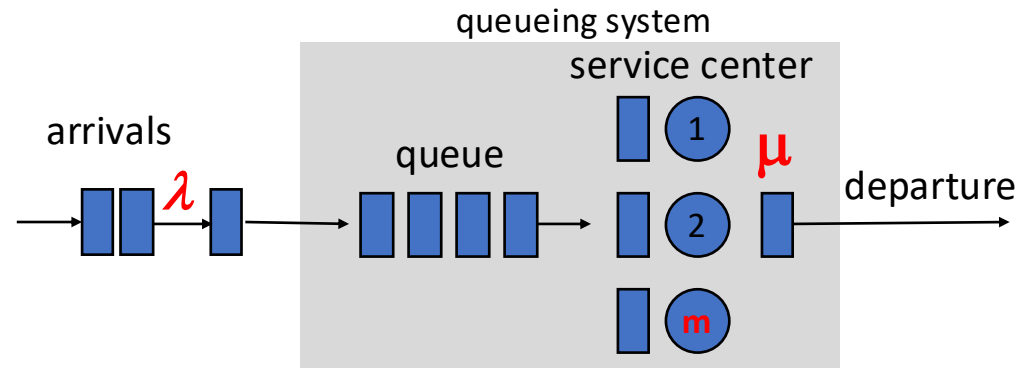


# CS5239 Computer System Performance Analysis

L5: Operational Analysis

# Rules for all Queues (L3)



Consider an arbitrary arrival stream of customers  $c_j, j=1,2,\dots$

- $t_j$  = arrival epoch of  $c_j$ ,  $t_0 \equiv 0$  &  $t_0 \leq t_1 \leq t_2 \dots$
- $\tau_j = t_j - t_{j-1}$  = inter-arrival time, i.e., time between two successive arrivals
- $\lambda$  = mean arrival rate, can be a function of the state of the system, e.g., depends on the number of jobs already in the system
- $s$  = service time per job
- $\mu$  = mean service rate per server, total service rate for  $m$  servers is  $m\mu$

# Little's Law (L3)

- $E[n]$  (mean number of jobs in the system) =  $E[n_q] + E[n_s]$  (1)  
where  $n_q$  = number of jobs waiting to receive service, and  $n_s$  = number of jobs receiving service
- $E[r]$  (mean response time in the system) =  $E[w] + E[s]$  (2)  
where  $w$  = waiting time of a job, and  $s$  = service time per job
- Little's Law:
  - mean number of jobs in system = mean arrival rate  $\times$  mean response time  
i.e.,  $E[n] = \lambda E[r]$  (3)
  - mean number of jobs in queue = mean arrival rate  $\times$  mean waiting time  
i.e.,  $E[n_q] = \lambda E[w]$  (4)

# Outline

- Operational Laws
  - Utilization Law
  - Forced Flow Law
  - Little's Law
  - General Response Time Law
  - Interactive Response Time Law
- Performance Bounds
  - Asymptotic Bounds
  - Balanced System Bounds
  - Examples: Bottleneck Analysis and Modification Analysis
- Summary

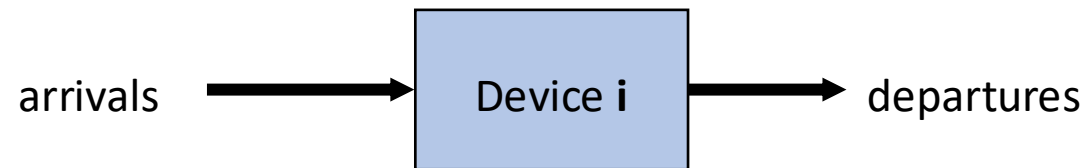


# Operational Laws (1)

- **Mean values** of performance metrics in queueing networks
- **No assumption** about the distribution of service times or inter-arrival times
- *Operational* means *directly measured*
- **Operational quantities** are directly measured during a finite observation period
- **Operational laws** are relationships among operational quantities

# Operational Laws (2)

Consider a black-box view of a device **i** (or service centre **i**):



If we observe the device for a finite amount of time **T** (observation period), we can measure:

1. number of job arrivals (**A<sub>i</sub>**)
2. busy time of device **i** during this period (**B<sub>i</sub>**)
3. number of job departures (**C<sub>i</sub>**)

# Operational Laws (3)

- Further operational quantities can be **derived**:

- (Internal) Arrival rate -  $\lambda_i = A_i / T$
- Throughput -  $X_i = C_i / T$
- Utilization -  $U_i = B_i / T$

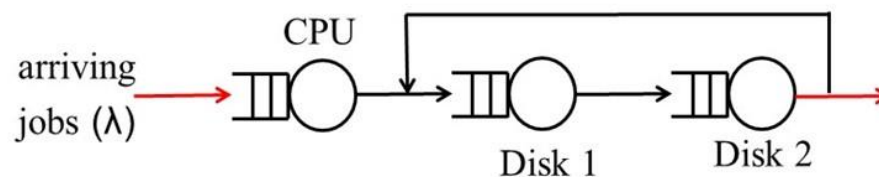
where  $i=0$  (system) and  $i > 0$  (devices)

- average service time **per job visit** to queue  $i$ ,

$$S_i = B_i / C_i$$

- average number of visits ( $V_i$ ) to device  $i$  by a job before it leaves the system

for a given job,  
disk1 and disk2  
are visited more  
than one time



jobs followed a  
predetermined route:  
CPU, disk 1, disk2,  
disk 1, disk 2 and out  
(leave system)

# Utilization Law ( $U_i = X_i S_i = \lambda_i S_i = X D_i$ )

- $U_i = B_i/T \times C_i/C_i = C_i/T \times B_i/C_i$   
 $= X_i S_i$

- Example: Packets leave a gateway at the rate of 125 packets/sec and the gateway takes an average of 2 msec to process each packet.

$$U_i = 125 * 0.002 = 0.25$$

- **Job flow balance assumption:** For a device  $i$ , number of job arrivals ( $A_i$ ) is equal to number of job completions ( $C_i$ ) in the observation period  $T$ .
- Assuming job flow balanced then  $A_i = C_i$  or  $A_i/T = C_i/T$  or  $U_i = \lambda_i S_i$

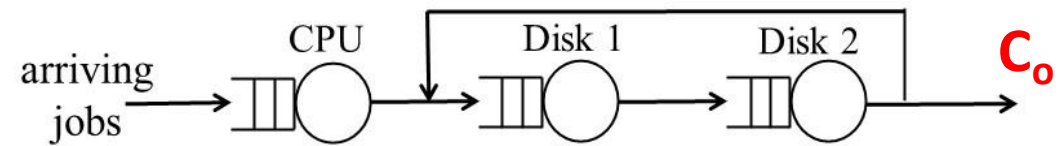
$$\text{Utilization Law: } U_i = X_i S_i = \lambda_i S_i$$

- The device with the largest utilization is the **bottleneck device**

# Visit Ratio or Number of Visits **per Job** to Device i

- Each job makes  $V_i$  requests for the  $i^{\text{th}}$  device in the system
- If job flow is balanced, the number of job arrivals is equal to the number of job completion ( $C_o$ )
- Thus, the number of jobs  $C_i$  visiting the  $i^{\text{th}}$  device is
$$C_i = C_o V_i \quad \text{or} \quad V_i = C_i / C_o$$
- $V_i$  is thus the ratio of visits to the  $i^{\text{th}}$  device and the outside link (job completed)
- For one job:  $V_i = C_i / 1 = \text{number of visits per job to device } i$

Example



$$C_{\text{disk1}} = C_o V_{\text{disk1}} \quad \text{or} \quad V_{\text{disk1}} = C_{\text{disk1}} / C_o$$

$$\text{For one job, } V_{\text{disk1}} = C_{\text{disk1}} / 1$$

For 10 jobs and  $C_{\text{disk1}} = 20$  then

$$V_{\text{disk1}} = 20 / 10 = 2$$

# Forced Flow Law ( $X_i = X V_i$ )

- Relates system throughput to individual device throughputs
- Each device satisfies the **job flow balance**, i.e.,  $A_i = C_i$
- Let  $X$  be the system throughput
  - for an **open network**, number of jobs leaving the system per unit time
  - for a **closed network**, number of jobs completing service
- Let  $C_0$  be the no. of jobs leaving the system in  $T$  then  $X = C_0/T$
- When job flow is balanced at device  $i$ ,  $V_i = C_i/C_0$  where  $V_i$  are called the **visit ratios**
- Throughput for device  $i$  is  $X_i = C_i/T \times C_0/C_0$  or  $X_i = X V_i$

# Forced Flow Law ( $X_i = XV_i$ )

- If  $D_i$  is the total **service demand** by a job at device  $i$  then

$$D_i = V_i S_i$$

- Next, combining forced flow law with utilization law, we have

$$U_i = X_i S_i = XV_i S_i = XD_i$$

- Let  $P_{ij}$  denotes the probability that a job on service completion at device  $i$  moves to device  $j$ ,  $P_{00} = 0$
- In a system with job flow balance, (assuming  $M$  devices in the system)

$$C_j = \sum_{i=0}^M C_i P_{ij}$$

- Dividing both sides by  $C_0$ , we get

$$V_j = \sum_{i=0}^M V_i P_{ij}$$

- This set of equations together with  $V_0 = 1$  are called **visit ratio equations**

# Example 1: Operational Laws

The accounting log for a time-shared computer system produces the following profile for user programs. Each program requires **5 seconds of CPU time** and makes **80 I/O requests to disk A** and **100 I/O requests to disk B**. The **average think time of the users was 18 seconds**. From the device specifications, it was determined that **disk A takes 50 milliseconds** to satisfy an I/O request and **disk B takes 30 milliseconds per request**. With **17 active terminals**, disk A throughput was observed to be **15.7 I/O requests per second**.

Determine the **system throughput** and **device utilizations**.

# Example 1: Operational Laws

## Input

$$D_{CPU} = 5 \text{ seconds}$$

$$V_A = 80$$

$$V_B = 100$$

$$Z = 18 \text{ seconds}$$

$$S_A = 0.05 \text{ second}$$

$$S_B = 0.03 \text{ second}$$

$$N = 17$$

$$X_A = 15.7 \text{ jobs/second}$$

System throughput,  $X = ?$

Device utilizations:

$$U_{CPU}, U_A, U_B = ?$$

(1)

From forced flow law :

$$X = \frac{X_A}{V_A} = \frac{15.7}{80}$$

$$\therefore X = 0.1963 \text{ job/second}$$

(2)

$$U_{CPU} = X D_{CPU} = 0.1963 * 5 = 98 \%$$

$$U_A = X_A S_A = 15.7 * 0.05 = 78.5 \%$$

$$U_B = X_B S_B = X_B * 0.03$$

$$= X V_B * 0.03$$

$$= 0.1963 * 100 * 0.03$$

$$\therefore U_B = 58.9 \%$$

# Little's Law

- Assume job flow balance
- If  $Q_i$  is the average queue size (including the jobs in service) and  $R_i$  is the average customer response time at queue  $i$ , then

$$Q_i = \lambda_i R_i = X_i R_i$$

# Example 1: Operational Laws

From example 1, if the queue lengths are:

$$Q_{CPU} = 8.88 \quad Q_A = 3.19 \quad Q_B = 1.4$$

**determine the device response times**

$$X_{CPU} = XV_{CPU} = 0.1963 \times \mathbf{181} = 35.48$$

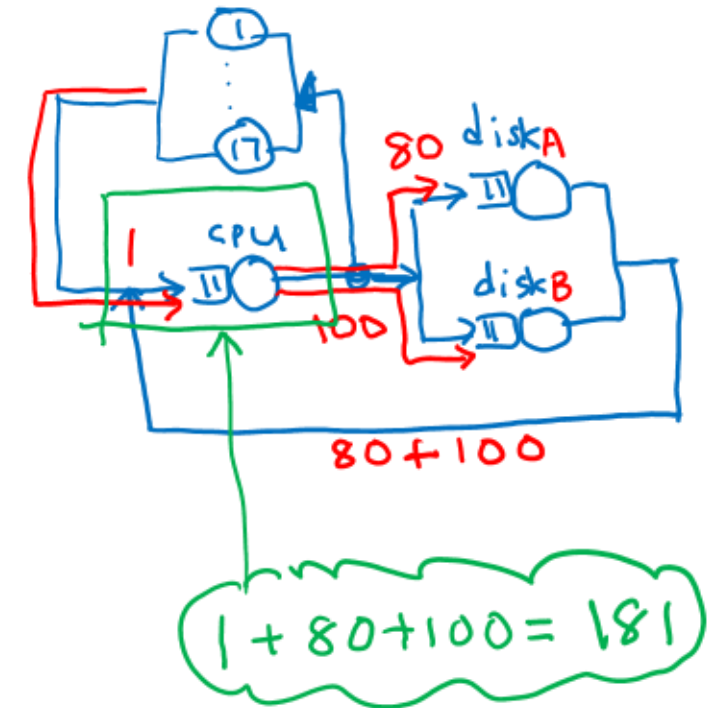
$$X_{CPU} = 35.53, \quad X_A = 15.7, \quad X_B = 19.6$$

Using Little's Law, the device response times are:

$$R_{CPU} = \frac{Q_{CPU}}{X_{CPU}} = \frac{8.88}{35.53} = 0.25 \text{ second}$$

$$R_A = \frac{Q_A}{X_A} = 0.203 \text{ second}$$

$$R_B = \frac{Q_B}{X_B} = 0.071 \text{ second}$$



# General Response Time Law

- Little's law can be applied to any part of the system, provided the job flow in that part is balanced
- Thus,  $Q = XR$  where  $Q$  is the average queue in the system and  $R$  is the average customer response time in the system
- Since  $Q = Q_1 + \dots + Q_M$

$$XR = \sum_{i=1}^M X_i R_i$$

- Dividing both sides by  $X$  and using the forced flow law:

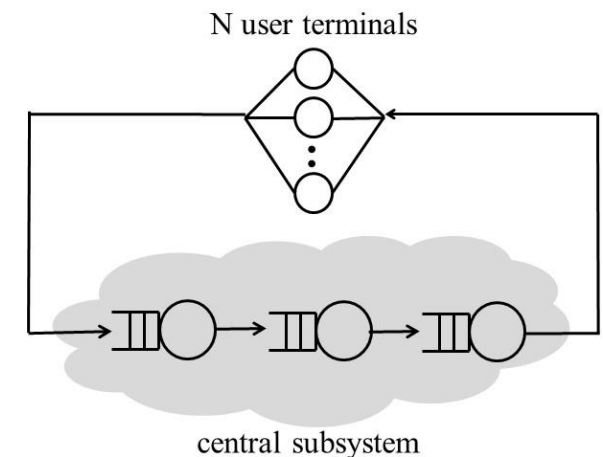
$$R = \sum_{i=1}^M V_i R_i$$

# Interactive and Batch systems

| Feature         | Interactive system                               | Batch system                                                    |
|-----------------|--------------------------------------------------|-----------------------------------------------------------------|
| Definition      | serves requests directly while users are waiting | collects jobs and runs them later, usually through a scheduler. |
| Main metric     | Response time                                    | Throughput and resource utilization                             |
| Workload        | Short, unpredictable requests                    | Long, predefined jobs                                           |
| Scheduling goal | Keep response time low                           | Keep resources busy and complete jobs efficiently               |
| Examples        | Shell command, web request, database query       | HPC simulation, rendering job, large data-processing task       |

# Interactive Response Time Law

- In an interactive system, a user generates a request and waits for the response; on getting the response, wait for some time (think time) and generates the next request
- Let **R** be the system response time and **Z** be the average think time
- Therefore, each user generates  $T/(R + Z)$  requests in the time period T
- For *N* users (terminals, closed system),
  - no. of jobs completed =  $N * T/(R + Z)$
  - system throughput  $X = (NT/(R + Z)) / T$   
 $= N/(R + Z)$  or  
 **$R = N/X - Z$**



# Example 1: Operational Laws

From example 1:

$$X = 0.1963, N = 17 \text{ and } Z = 18$$

$$R = 17/0.1963 - 18$$

$$= \mathbf{68.6 \text{ seconds}}$$

## Response Time and Service Time (one server, FCFS)

- $R_i$  is the response time at device  $i$  and can be decomposed into the time spent in service ( $S_i$ ) and time spent queueing:
  - If there are  $N_i$  other jobs **waiting** at device  $i$ ,  $N_i$  other jobs have to finish service before the current job can get service

$$R_i = N_i S_i + S_i = R_i X_i S_i + S_i = R_i * U_i + S_i$$

$$R_i = \frac{S_i}{1-U_i}$$

# Operational Analysis Summary

$i=0$  (system) and  $i > 0$  (devices)

- Arrival rate (arrivals):  $\lambda_i = \frac{A_i}{T}$
- Throughput (completed jobs):  $X_i = \frac{C_i}{T}$
- Utilization (busy time):  $U_i = \frac{B_i}{T}$
- Job flow balance:  $A_i = C_i$  or  $\lambda_i = X_i$
- Average service time per job visit to queue  $i$ :  $S_i = \frac{B_i}{C_i}$
- Average number of visits to device  $i$  by a job before it leaves the system, visit ratio:  $V_i = \frac{C_i}{C_0}$
- Total service demand by a job at device  $i$ :  $D_i = V_i S_i$
- Response time:  $R_i$
- Number of jobs at queue  $i$ , including any job currently in service:  $Q_i$

## 1. Utilization Law

$$U_i = X_i S_i = \lambda_i S_i = X D_i$$

## 2. Forced Flow Law

$$X_i = X V_i$$

## 3. Little's Law

$$Q_i = \lambda_i R_i = X_i R_i \text{ (device } i)$$

$$Q = X R \text{ (system)}$$

## 4. General Response Time Law

$$R = \sum_{i=1}^M V_i R_i$$

## 5. Interactive Response Time Law

$$R = \frac{N}{X} - Z$$

# Outline

- Operational Laws
  - Utilization Law
  - Forced Flow Law
  - Little's Law
  - General Response Time Law
  - Interactive Response Time Law
- Performance Bounds
  - Asymptotic Bounds
  - Balanced System Bounds
  - Examples: Bottleneck Analysis and Modification Analysis
- Summary



# Performance Bounds

- Why bounding analysis?
  - Provides insight on **primary factors** affecting performance, e.g., identify and quantify the degree of bottleneck
  - Provides a first cut modeling technique **to eliminate inadequate alternatives** at the early stage of a study
- **Asymptotic bounds**: determine *upper* and *lower bounds*, e.g., system throughput and response times as functions of system workload intensity
- **Balanced bounds**: performance when device demands are balanced, i.e., **no bottleneck**

# Asymptotic Bounds

- Examples:
  - **Optimistic (throughput)** and **pessimistic (response time)** bounds
  - How – considers extreme conditions of light and heavy loads
- Validity of the bounds depends on ONLY one single assumption:

Service demand of a customer at a center does not depend on how many other customers currently are in the system, or at which service centers they are located.

# Asymptotic Bounds

- From forced flow law, device utilizations are proportional to total service demands:  $U_i \propto D_i$

$$U_i = X_i S_i = \lambda_i S_i = X D_i$$

- Thus, device with the **highest** total service demand  $D_i$  has the highest utilization and is the **bottleneck device**
- If device **b** is the bottleneck, then  $D_b = D_{\max}$  is the highest among  $D_1, D_2, \dots, D_m$
- $D$  is total (sum of) service demands on all devices except terminals

# Asymptotic Bounds

- Based on the following observations:
  1. Utilization of any device cannot **exceed 1**. This puts a limit on the *maximum* obtainable throughput - upper bound (optimistic throughput)
  2. Response time of the system with N users cannot be less than a system with just one user. This puts a limit on the *minimum* response time – lower bound (optimistic response)
  3. Use interactive response time formula

$$R = \frac{N}{X} - Z$$

## Optimistic Bound Derivation:

- For bottleneck device **b** we have:

$$U_b = XD_{\max} \text{ (utilization law)}$$

- since  $U_b \leq 1$ ,  $XD_{\max} \leq 1$ , so  $X \leq 1/D_{\max}$  (1)
- With **one job**, there is zero waiting time at all queues,

$$R = \sum_{i=1}^M D_i = D$$

- With **more than one user**

$$R \geq D \quad (2)$$

- combining the interactive response time law ( $R = \frac{N}{X} - Z$ ) with inequality (1), we get

$$R(N) \geq ND_{\max} - Z \quad (3)$$

- combining (2) & (3) - bound,

$$R(N) \geq \max(D, ND_{\max} - Z) \quad (4)$$

- combining the interactive response time law ( $R = \frac{N}{X} - Z$ ) with inequality (2), we get

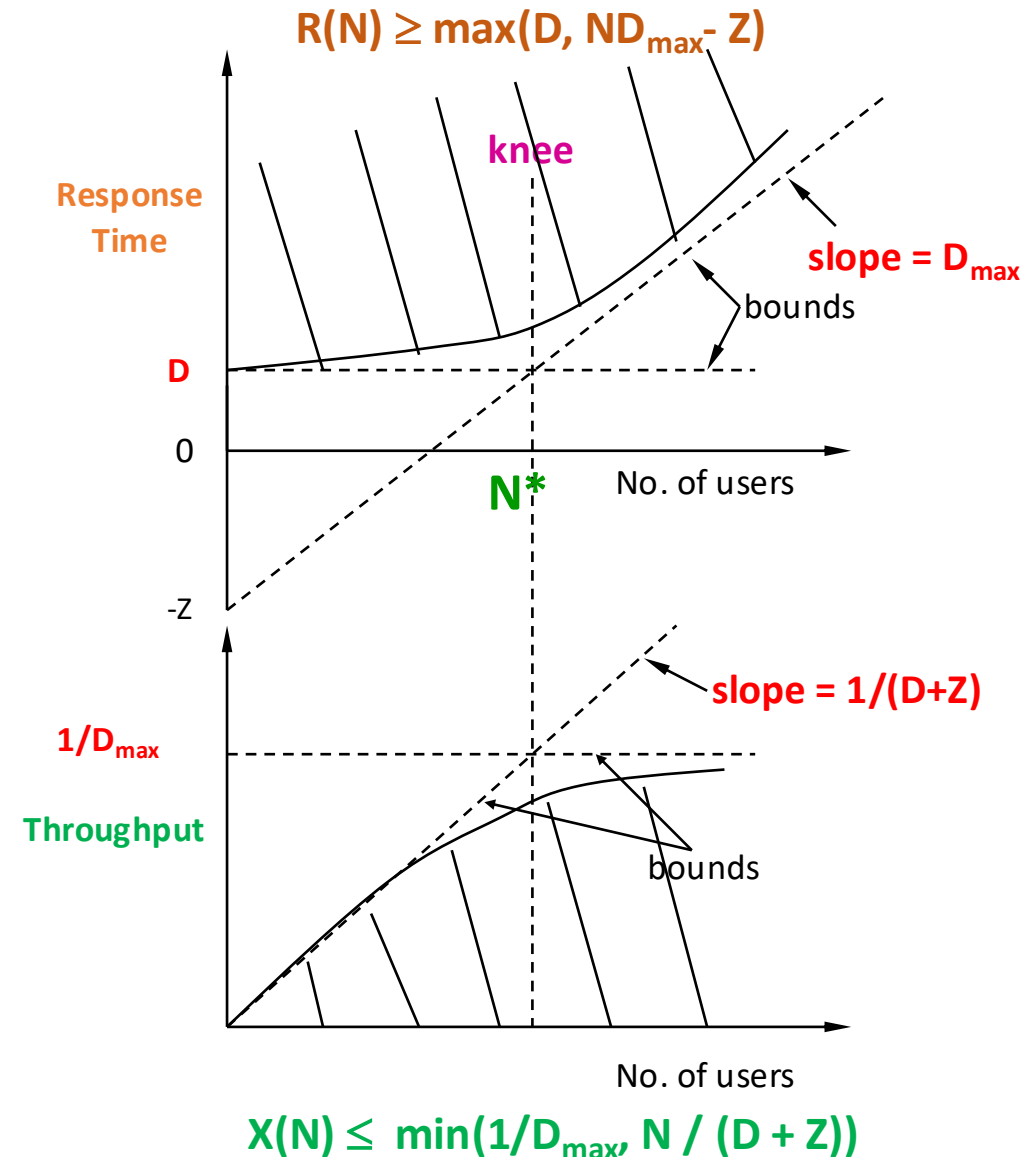
$$X(N) \leq N / (D + Z) \quad (5)$$

- combining (1) & (5) - bound,

$$X(N) \leq \min(1/D_{\max}, N / (D + Z)) \quad (6)$$

- For batch systems,  $Z = 0$

## Typical asymptotic bounds



# Asymptotic Bounds

- $D = ND_{\max} - Z$   
or  $N^* = (D+Z) / D_{\max}$  (number of users at the knee)

For  $N > N^*$  there is significant congestion in the system

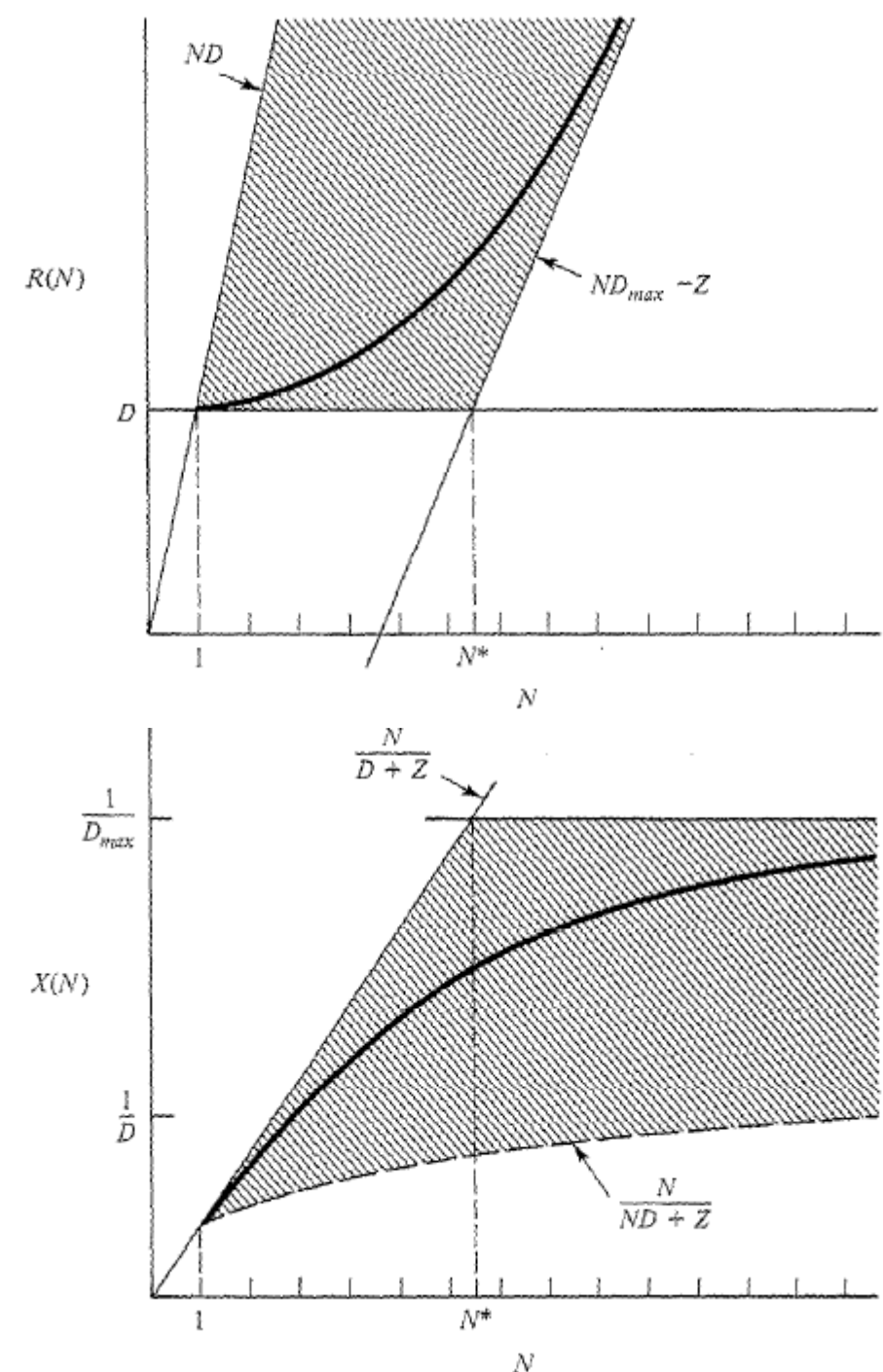
**For  $N < N^*$ , as  $N$  increases, response time does not increase significantly, and throughput increases very fast**

# Pessimistic Bounds

- The smallest possible throughput occurs when each additional customer is forced to queue behind all other customers already in the system.
  - $D$  time units are spent in service
  - $(N - 1)D$  time units are spent queued behind other customers
  - $Z$  time units are spent thinking

The throughput of each customer is  $1/(ND + Z)$

System throughput is  $N/(ND + Z)$ .



# Asymptotic Bounds (Summary)

## ASYMPTOTIC BOUNDS

System throughput:

$$N/(ND+Z) \leq \mathbf{X(N)} \leq \min \{1/D_{\max}, N/(D + Z)\}$$

System response time:

$$\max \{ D, ND_{\max} - Z \} \leq \mathbf{R(N)} \leq ND$$

Number of users at the knee:

$$\mathbf{N^*} = (D+Z)/D_{\max}$$

# Example: Asymptotic Bounds (1)

From example 1:

$$D_{\text{CPU}} = 5, D_A = 4, D_B = 3, Z=18$$

$$D = D_{\text{CPU}} + D_A + D_B = 12$$

$$D_{\text{max}} = D_{\text{CPU}} = 5$$

The asymptotic bounds are:

$$R(N) \geq \max\{D, ND_{\text{max}} - Z\} = \max\{12, 5N - 18\}$$

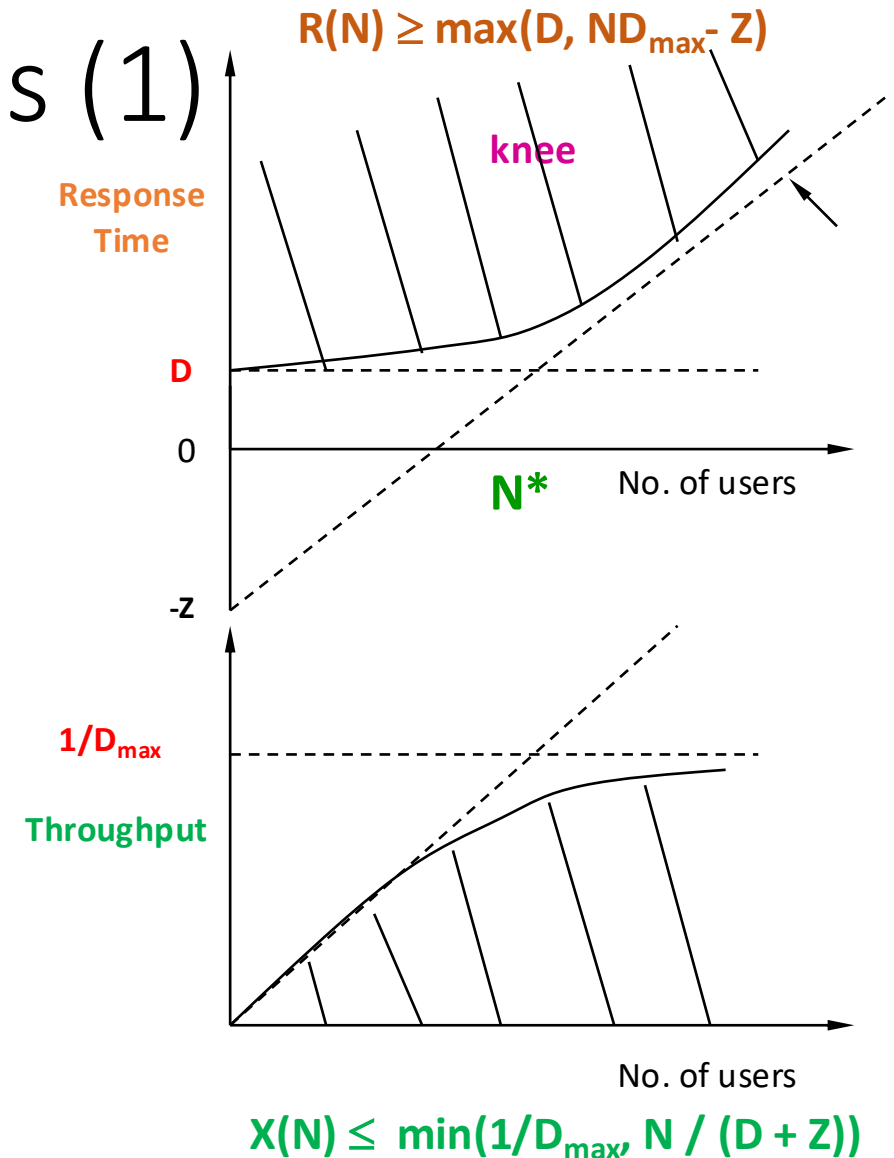
$$X(N) \leq \min\left\{\frac{N}{D+Z}, \frac{1}{D_{\text{max}}}\right\} = \min\left\{\frac{N}{30}, \frac{1}{5}\right\}$$

Knee occurs at:

$$12 = 5N - 18$$

$$\therefore N^* = 6$$

If there are more than 6 users, there will certainly be congestion in the system.



# Example: Asymptotic Bounds (2)

- How many terminals can be supported if response time has to be kept below 100 seconds ?

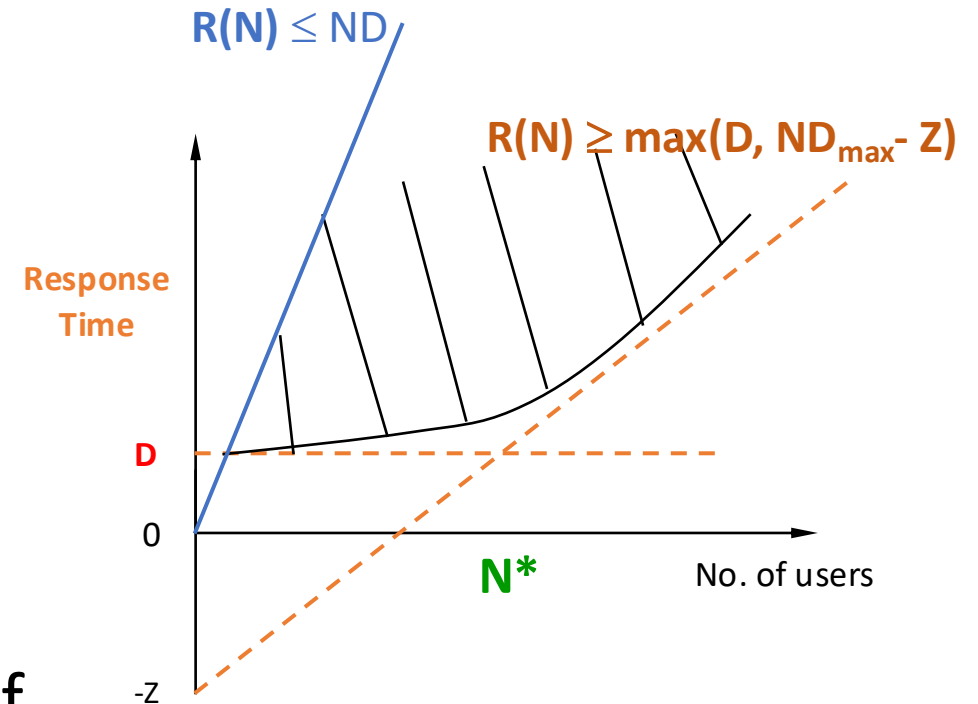
Since

$$R(N) \leq ND \leq 100$$

$$5N \leq 100$$

$$N \leq 20$$

System cannot support more than 20 users if required response time is less than 100 seconds



# Balanced System Bounds

- Tighter bounds (complements asymptotic bounds)
- Balanced system:  $D_1 = D_2 = D_3 \dots = D_k$
- Batch Workload

Given that utilization at a service centre:

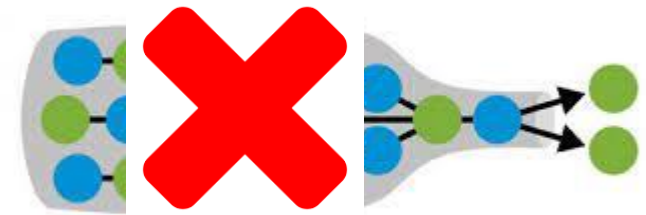
$$U_k(N) = N/(N+K-1)$$

where  $N$  = number of jobs,  $K$  = number of service centers,

By utilization law ( $U_i = X D_i$ ):

$$X(N) = U_k/D_k = N/(N+K-1) \times 1/D_k$$

where  $D_k$  = service demand at every center



# Balanced System Bounds

Let  $D_{\max}$ ,  $D_{\text{ave}}$  and  $D_{\min}$  denote maximum, average and minimum service demands respectively

Let's bound system throughput using throughputs of two balanced systems:

- a. one with service demand  $D_{\min}$  at every service center
- b. the other with service demand  $D_{\max}$  at every center

Thus,

$$N/(N+K-1) \times 1/D_{\max} \leq \mathbf{X(N)} \leq N/(N+K-1) \times 1/D_{\min} \quad (7)$$

and response time:

$$(N+K-1) D_{\min} \leq \mathbf{R(N)} \leq (N+K-1) D_{\max} \quad (8)$$

# Balanced System Bounds

Of all systems, the one with the highest throughput is the one in which all service demands are equal, i.e.,  $D_k = D/K$ ,  $k=1,2,\dots,K$ .

Thus, *optimistic* bounds are:

$$X(N) \leq N/(N+K-1) \times 1/D_{ave} = N/(D+(N-1)D_{ave}) \quad \text{- from (7)}$$

and

$$D + (N-1)D_{ave} \leq R(N) \quad \text{- from (8)}$$

Beyond point  $N^+$  (intersects of balanced system bound and heavy load of asymptotic bound), the balanced system bound coincides with the asymptotic bound. Thus, service center with the lowest throughput has  $D/D_{max}$  centers with demand  $D_{max}$ .

The *pessimistic* bounds are:

$$N/(N+(D/D_{max})-1) \times 1/D_{max} = N/(D+(N-1)D_{max}) \leq X(N) \quad \text{- from (7)}$$

and

$$R(N) \leq D + (N-1)D_{max} \quad \text{- from (8)}$$

# Balanced System Bounds

Of all systems, the one with the highest throughput is the one in which all service demands are equal, i.e.,  $D_k = D/K$ ,  $k=1,2,\dots,K$ .

Thus, *optimistic* bounds are:

## Balance System Bounds

and

$$N/(D+(N-1)D_{\max}) \leq X(N) \leq \min(1/D_{\max}, N/(D+(N-1)D_{\text{ave}}))$$

Beyond  
balance  
through

$$\max(ND_{\max}, D + (N-1)D_{\text{ave}}) \leq R(N) \leq D + (N-1)D_{\max}$$

bound), the  
the lowest

The pes

$$N/(N+(D/D_{\max})-1) \times 1/D_{\max} = N/(D+(N-1)D_{\max}) \leq X(N) \quad \text{- from 7}$$

and

$$R(N) \leq D + (N-1)D_{\max} \quad \text{- from 8}$$

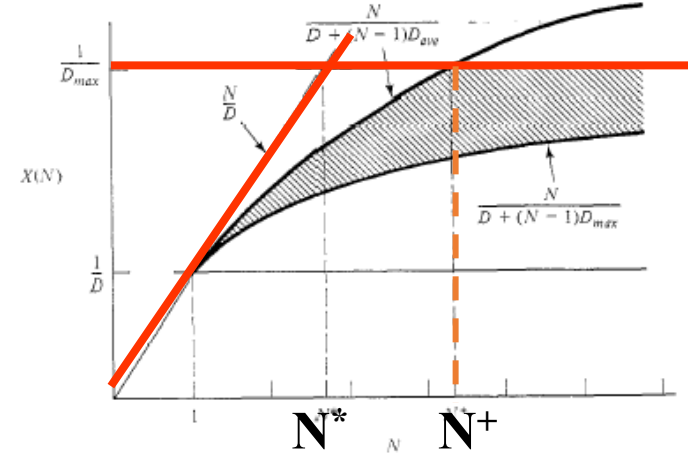
# Balanced System Bounds

|   | workload type | bounds                                                                                                                                                     |
|---|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| X | batch         | $\frac{N}{D + (N-1)D_{max}} \leq X(N)$ $\leq \min \left( \frac{1}{D_{max}}, \frac{N}{D + (N-1)D_{ave}} \right)$                                            |
|   | terminal      | $\frac{N}{D + Z + \frac{(N-1)D_{max}}{1 + Z/(ND)}} \leq X(N)$ $\leq \min \left( \frac{1}{D_{max}}, \frac{N}{D + Z + \frac{(N-1)D_{ave}}{1 + Z/D}} \right)$ |
|   | transaction   | $X(\lambda) \leq 1 / D_{max}$                                                                                                                              |
| R | batch         | $\max (ND_{max}, D + (N-1)D_{ave}) \leq R(N)$ $\leq D + (N-1)D_{max}$                                                                                      |
|   | terminal      | $\max (ND_{max} - Z, D + \frac{(N-1)D_{ave}}{1 + Z/D}) \leq R(N)$ $\leq D + \frac{(N-1)D_{max}}{1 + Z/(ND)}$                                               |
|   | transaction   | $\frac{D}{1 - \lambda D_{ave}} \leq R(\lambda) \leq \frac{D}{1 - \lambda D_{max}}$                                                                         |

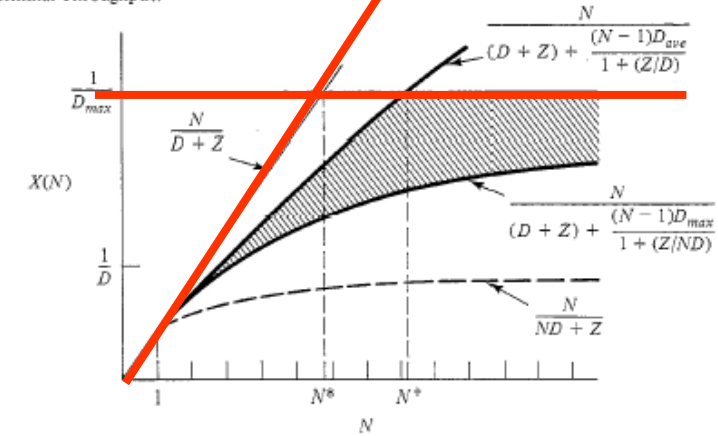
# Balanced System Bounds

$$N/(D+(N-1)D_{\max}) \leq \mathbf{X(N)} \leq \min(1/D_{\max}, N/(D+(N-1)D_{\text{ave}}))$$

Batch Throughput:

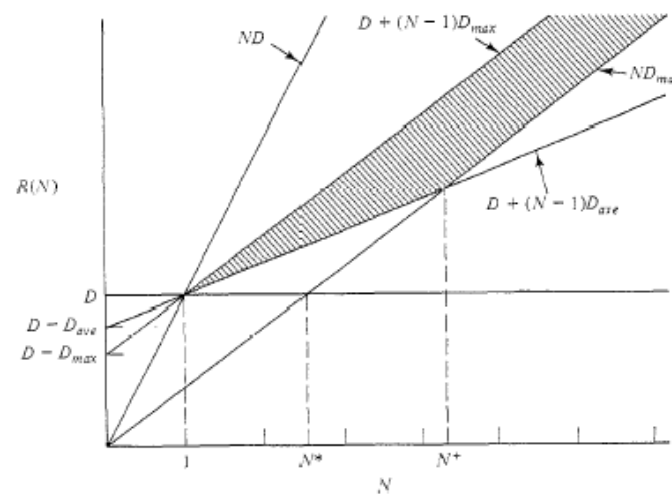


Terminal Throughput:

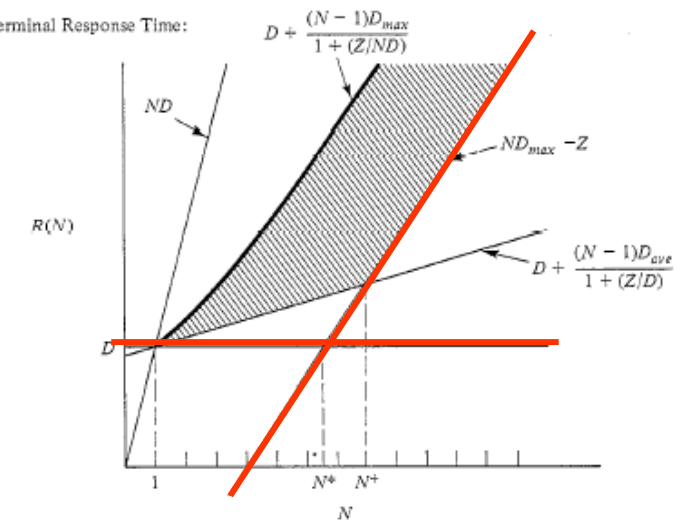


$$\max(ND_{\max}, D + (N-1)D_{\text{ave}}) \leq \mathbf{R(N)} \leq D + (N-1)D_{\max}$$

Batch Response Time:



Terminal Response Time:



# Bounds (Summary)

## ASYMPTOTIC BOUNDS

System throughput:

$$N/(ND+Z) \leq \mathbf{X(N)} \leq \min \{1/D_{\max}, N/(D + Z)\}$$

System response time:

$$\max \{ D, ND_{\max} - Z \} \leq \mathbf{R(N)} \leq ND$$

Number of users at the knee:

$$\mathbf{N^*} = (D+Z)/D_{\max}$$

## BALANCE SYSTEM BOUNDS

$$N/(D+(N-1)D_{\max}) \leq \mathbf{X(N)} \leq \min (1/D_{\max}, N/(D+(N-1)D_{\text{ave}}))$$

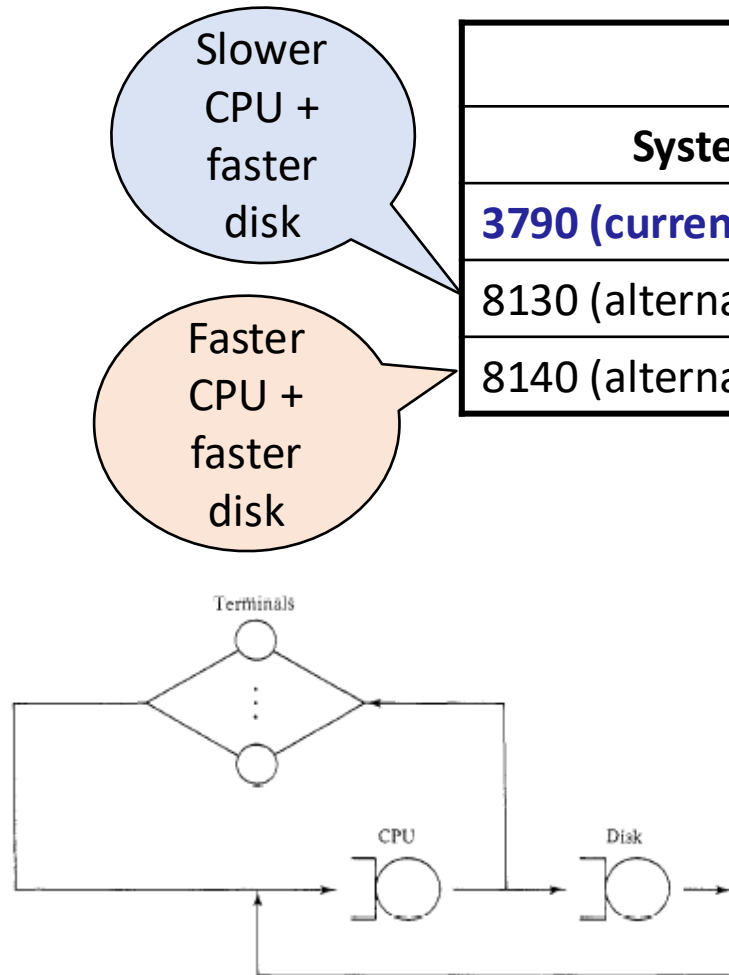
$$\max(D_{\max}, D + (N-1)D_{\text{ave}}) \leq \mathbf{R(N)} \leq D + (N-1)D_{\max}$$

## Example 2: Bottleneck Analysis

An insurance company has twenty geographically distributed sites based on IBM 3790s. The company is considering a system upgrade and *two alternatives*, IBM 8130s and 8140s, are proposed. After discussion with the vendors, the company believes that the 8130s would improve performance by a factor of 1.5 to 2 over the 3790s while the 8140s would improve performance by a factor of 2 to 3.5.

- 8130 processor ([alternative 1](#)) slightly slower than current and 8140 ([alternative 2](#)) is 1.5 times faster
- Both included a disk that is substantially faster

# Example 2: Bottleneck Analysis



Slower  
CPU +  
faster  
disk

Faster  
CPU +  
faster  
disk

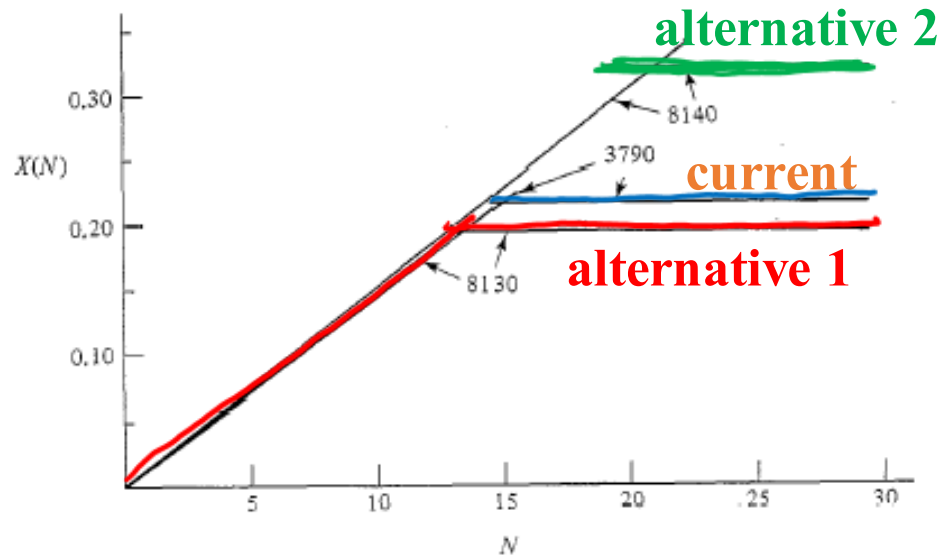
| System                | Service demands (seconds) |      |
|-----------------------|---------------------------|------|
|                       | CPU                       | Disk |
| <b>3790 (current)</b> | 4.6                       | 4.0  |
| 8130 (alternative 1)  | <b>5.1</b>                | 1.9  |
| 8140 (alternative 2)  | 3.1                       | 1.9  |

The parameters are:

- K, the number of service centers (2)
- $D_{\max}$ , the largest service demand (4.6 seconds for the 3790, **5.1 for the 8130**, and 3.1 for the 8140)
- D, the sum of the service demands (8.6, 7.0, and 5.0, respectively)
- type of the customer class (terminal)
- Z, average think time (estimated at 60 seconds)

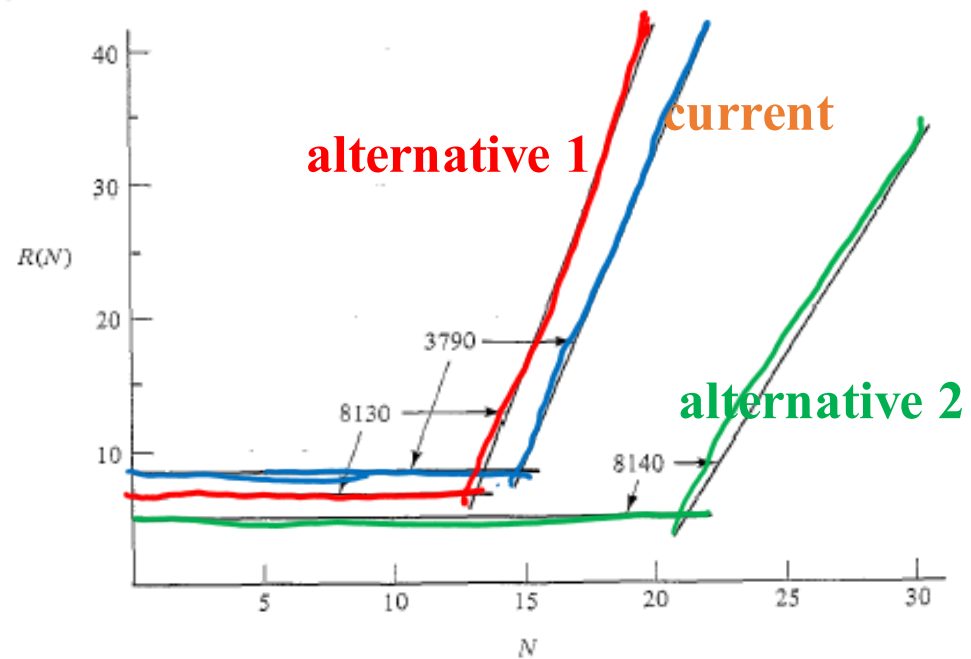
# Example 2: Bottleneck Analysis

Throughput:



|                       | Service demands (seconds) |            |
|-----------------------|---------------------------|------------|
| System                | CPU                       | Disk       |
| <b>3790 (current)</b> | <b>4.6</b>                | <b>4.0</b> |
| 8130 (alternative 1)  | <b>5.1</b>                | 1.9        |
| 8140 (alternative 2)  | <b>3.1</b>                | 1.9        |

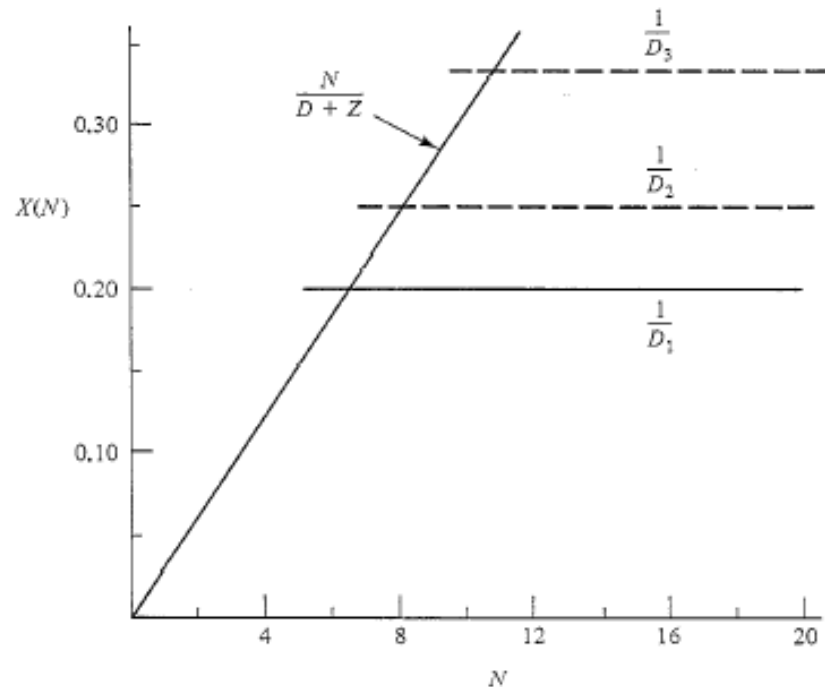
Response Time:



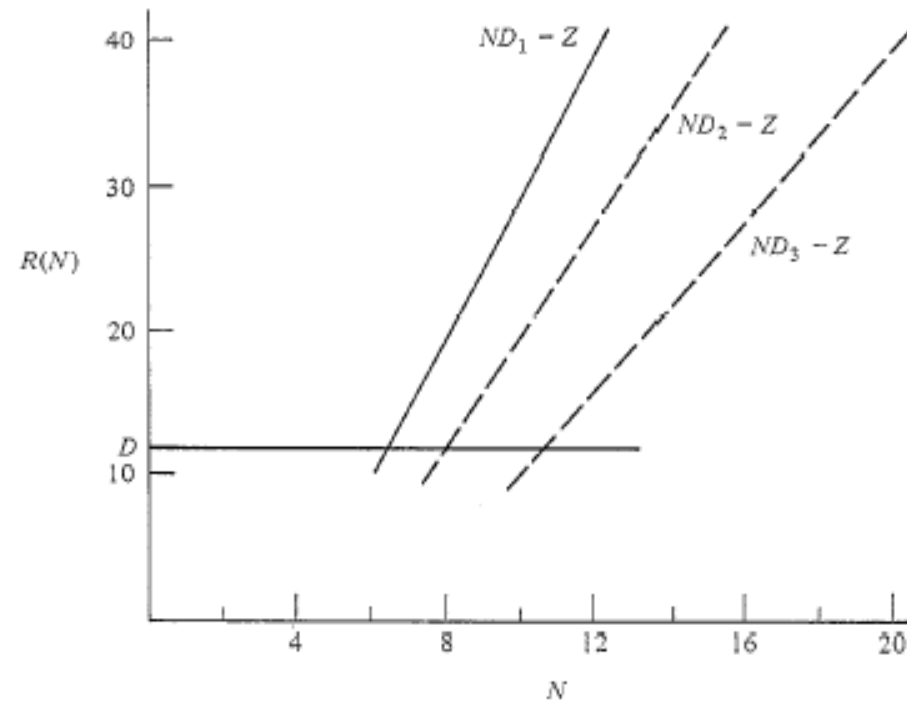
# Example 3: Bottleneck Analysis

- Consider a model with three service centers ( $K=3$ ) and a terminal workload with average think time of 15 seconds ( $Z=15$ ) and service demands of 5, 4, and 3 seconds at the centers ( $D_1=5$ ,  $D_2=4$ , and  $D_3=3$ )

Throughput:



Response Time:

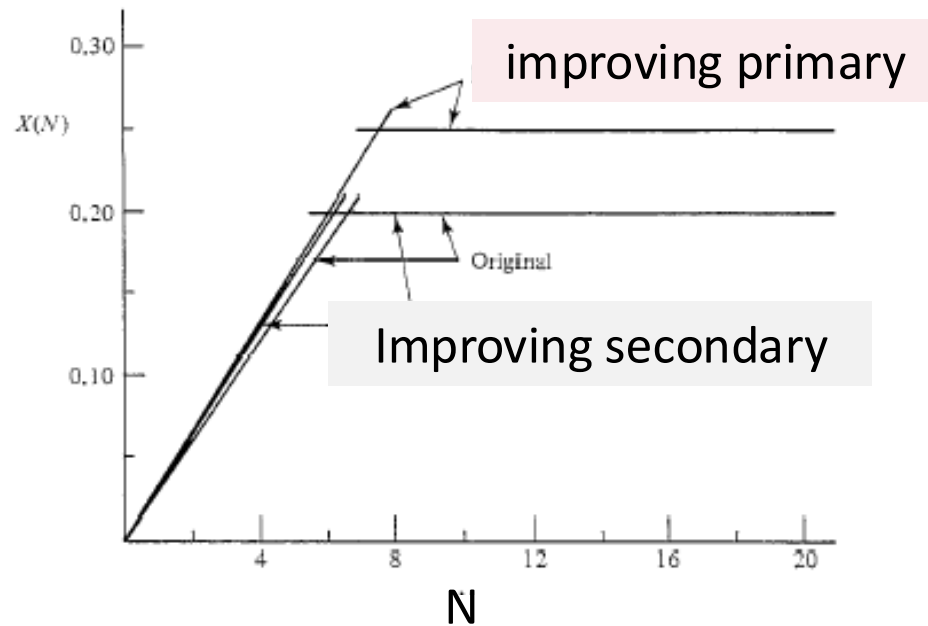


# Example 3: Relative Effects

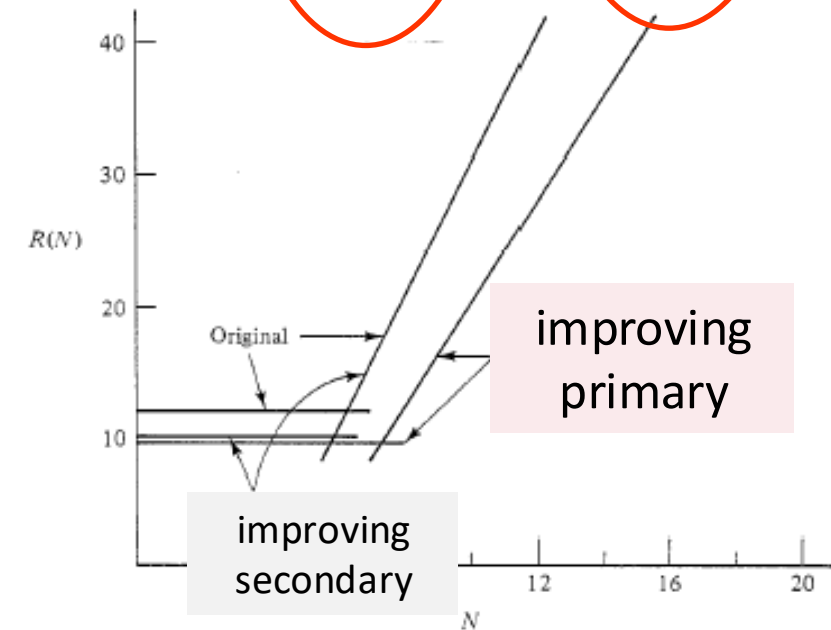
Primary and secondary bottleneck?  
improve **primary** bottleneck first!

|       | Original | Primary | Secondary |
|-------|----------|---------|-----------|
| $D_1$ | 5        | 2.5     | 5         |
| $D_2$ | 4        | 4       | 2         |
| $D_3$ | 3        | 3       | 3         |

Throughput:



Response Time:



# Example 4: Modification Analysis

Consider an interactive system with the following obtained using measurement:

$T = 900$  seconds length of the measurement interval

$B_1 = 400$  seconds CPU busy

$B_2 = 100$  seconds slow disk busy

$B_3 = 600$  seconds fast disk busy

$C = 200$  jobs completed jobs

$C_2 = 2,000$  slow disk operations

$C_3 = 20,000$  fast disk operations

$Z = 15$  seconds think time

- Throughput (completed jobs):  $X_i = \frac{C_i}{T}$

- Utilization (busy time):  $U_i = \frac{B_i}{T}$

- Visit ratio:  $V_i = \frac{C_i}{C_0}$

- Total service demand by a job at device  $i$ :  $D_i = V_i S_i$

- Utilization Law:  $U_i = X D_i$

- Forced Flow Law:  $X_i = X V_i$

Four modifications:

1. Replace the CPU with one that is *twice* as fast, i.e.,  $D_1 \leftarrow 1$
2. Shift some files from the faster disk to the slower disk, *balancing* their demands
3. Add a *second fast disk* (center 4) to handle half the load of the busier existing disk
4. *All three changes made together: the faster CPU and a balanced load across two fast disks and one slow disk*

# Example 4: Modification Analysis

## Service Demand (D)

CPU ( $D_1$ )

Slow Disk ( $D_2$ )

Fast Disk ( $D_3$ )

Original

(1)

(2)

(3)

All

2

1

2

2

1

0.5

0.5

2.06

0.5

1.27

3.0

3.0

2.06

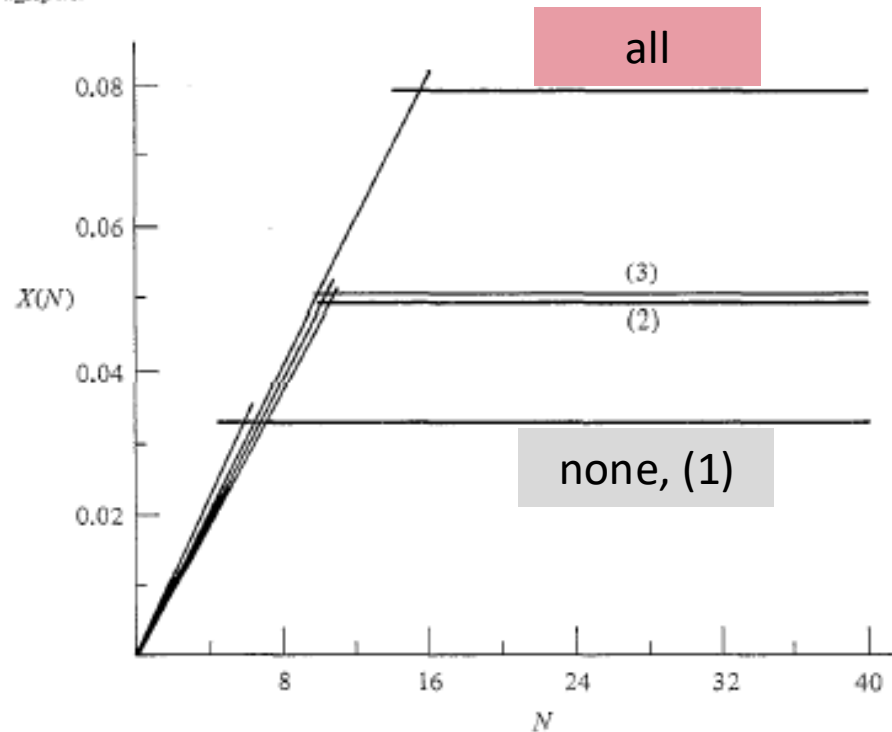
1.5

1.27

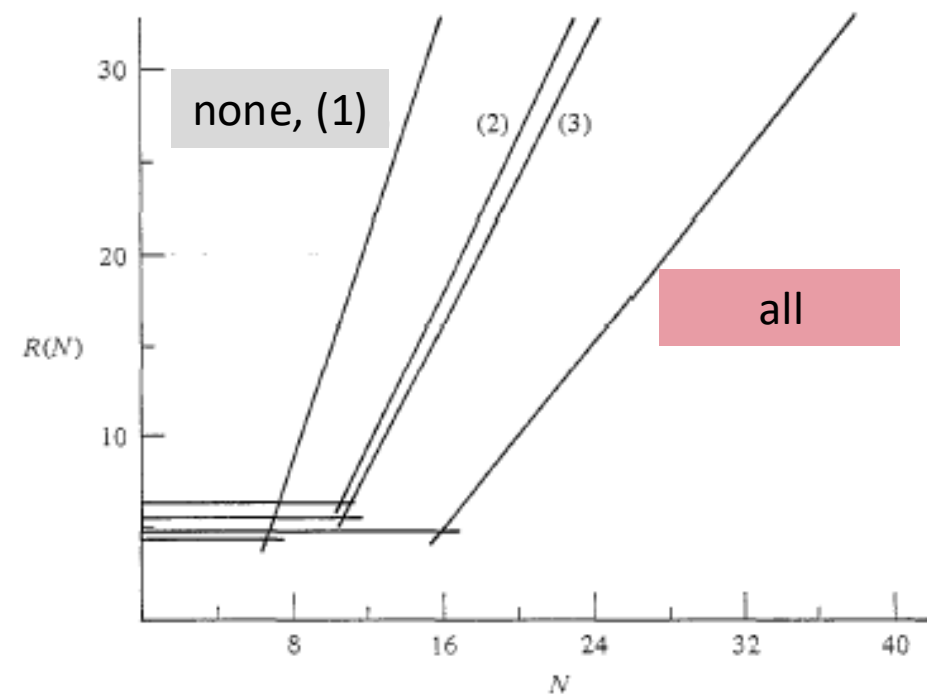
1.5

1.27

Throughput:



Response Time:



# Summary

- Mean value analysis
- 5 performance laws
- Performance bounds – asymptotic and balanced system bounds, bottleneck and modification analyses
- References:
  - [Jain 91] - chapter 33
  - Quantitative System Performance. Computer System Analysis Using Queueing Network Models – by Edward D. Lazowska et al. - chapters 3, 4 & 5 (<https://homes.cs.washington.edu/~lazowska/qsp/>)